

Yinan Dong

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EDUCATION

University of Michigan, Ann Arbor, MI

Bachelor of Engineering in Data Science

Expected Graduation: May 2026

GPA: 3.93 / 4.00

Shanghai Jiao Tong University, Shanghai, China

Bachelor of Engineering in Mechanical Engineering

Expected Graduation: Aug 2026

Core GPA: 3.50 / 4.00

PUBLICATIONS

- **Y. Dong**, Z. Xu, T. Lazouski, S. Teng, and M. Ghaffari, “Online Learning–Enhanced Lie Algebraic MPC for Robust Trajectory Tracking of Autonomous Surface Vehicles,” *IEEE Robotics and Automation Letters (RA-L)*, under review, 2025. [arXiv:2511.18683](https://arxiv.org/abs/2511.18683)

RESEARCH EXPERIENCE

University of Michigan – Curly Lab (Prof. Maani Ghaffari)

Oct 2024 – Present

- Developed an online learning–control algorithm for disturbance compensation in ASVs.
- Optimized Lie-algebraic MPC via CasADi, reducing loop time from 40 ms to 15 ms.
- Modeled ASV hardware and dynamics; integrated models into a simulation testbed.
- Designed and fabricated an onboard sensor suite (LiDAR, camera, IMU, Jetson).

University of Michigan – Z Lab (Prof. Y. Z.)

Jan 2025 – Present

- Designed hardware for swarm vibration robots and estimated their dynamics.
- Implemented swarm motion-planning algorithms on physical robotic platforms.

Shanghai Jiao Tong University – VEX Robotics Team

Sep 2022 – Jul 2024

- Conducted CAD design, 3D printing, and metal fabrication for autonomous robotic vehicles.
- Tuned robot controllers for autonomous task completion in competition environments.

HONORS & AWARDS

SJTU Undergraduate Excellent Scholarship (2022–2023)

Nov 2023

Champion & Skills Champion, VEX Robotics Asia Open Finals

Feb 2024

Inspire Award, VEX Robotics World Championship

May 2024

SJTU Undergraduate Excellent Scholarship (2023–2024)

Nov 2024

Dean’s List, University of Michigan FA 2024

Feb 2025

Dean’s List, University of Michigan WN 2025

Apr 2025

EMPLOYMENT HISTORY

University of Michigan – Research Assistant

May 2025 – Aug 2025

- Conducted summer research at the Curly Lab focusing on robust control for ASVs.

Lacuum – Hardware Engineer

March 2025 – Present

- Performed mechanical design for vessels and payload integration.
- Built detailed CAD models and carried out rapid prototyping for validation.

Shanghai Jiao Tong University – Student Assistant

Sep 2023 – Aug 2024

- Instructed students in safe operation of 3D printers, saws, and other fabrication tools.
- Supervised lab safety and assisted with project prototyping and maintenance.

SKILLS

- **Programming:** Python, C++, C, HTML, CSS, JavaScript, R, SQL
- **Tools:** ROS 2, IsaacSim, SolidWorks, Gazebo, Git, LaTeX, CasADi